



Institute of Advanced Research
The University for Innovation

INSTITUTE OF ADVANCED RESEARCH (IAR)

The University for Innovation | Gandhinagar, Gujarat

FIRST PROGRESS WORK PRESENTATION • 24TH SEPTEMBER 2026 • 10 MARKS

E-COMMERCE CUSTOMER LIFECYCLE & RFM SEGMENTATION DASHBOARD

An Applied Machine Learning & Analytics Engineering System for Behavioral Cohort Optimization

PROJECT SCHOLAR

Candidate Name: Mr. Deep Koshiya
Program: B.Sc. in AI & Data Science
Semester: Semester V (5th Semester)
Academic Year: 2026 – 2027
Course: Project - 1 (CE-502 / PRJ-701)

EVALUATION & MILESTONE SCOPE

Milestone: First Progress Work Evaluation
Evaluation Date: 24th September 2026 (10 AM - 5 PM)
Marks Allotted: 10 Marks (Curriculum Deliverable)
Syllabus Scope: Sections 1.1, 1.2, 1.3, 2.1, 2.2
Supervision: Assigned Project Supervisor (IAR)

Problem Definition & Commercial Context

1. Commercial Retention Dilemma

- **Surging Acquisition Costs:** Digital advertising CAC has increased over 60%, making organic customer retention the primary driver of profitability.
- **Homogenous Marketing:** Treating all shoppers equally leads to margin-eroding discounts for customers who would have purchased anyway.
- **Undetected Churn Risk:** Historically high-value accounts slip into dormancy silently without early churn detection triggers.
- **Missed Lifetime Value:** E-commerce retailers fail to identify and nurture accounts transitioning into Champions.

2. Limitations of OLTP Systems

- **Order-Centric Storage:** Traditional databases record individual shopping transactions, not longitudinal customer behavioral lifecycles.
- **Data Quality Deficits:** Event logs contain guest checkouts lacking CustomerID, cancellations, and administrative line-items.
- **Distribution Anomalies:** Extreme wholesale B2B distributor purchases contaminate standard consumer metrics.
- **No Analytical Layer:** Decision-makers lack an automated transformation pipeline bridging raw order rows with strategic marketing actions.

3. Targeted Project Solution

- **Automated Refinery:** Cleanses raw transactions, isolates reversals into an audit ledger, and caps outliers via Tukey IQR fences.
- **Vectorized RFM Feature Store:** Aggregates transactions into Recency, Frequency, Monetary spend, and Average Order Value (AOV).
- **Skew-Resistant ML Clustering:** Applies log1p and StandardScaler before K-Means modeling, validated via Silhouette diagnostics.
- **Executive BI Dashboard:** Interactive web UI with portfolio KPI scorecards, spatial scatterplots, and exportable CRM cohorts.

Requirement Specifications (Functional & Non-Functional)

Functional Requirements Specifications (FR-01 to FR-06)

- FR-01: Ingestion & Schema Typing:** Ingests raw CSV streams, validates typing (InvoiceDate to datetime64[ns], Quantity/UnitPrice to floats), and sanitizes SKU descriptions.
- FR-02: Missing Entity Resolution:** Detects and filters guest checkouts missing CustomerID to protect longitudinal cohort integrity.
- FR-03: Reverse Logistics Isolation:** Routes cancellations ('^C') and credit adjustments to an isolated audit store (fact_reversals_audit.csv) without distorting purchase counts.
- FR-04: Tukey IQR Outlier Capping:** Applies fences [Q1 - 1.5*IQR, Q3 + 1.5*IQR] on Quantity and Price, neutralizing bulk distributor volume spikes.
- FR-05: Vectorized RFM Derivation:** Computes Recency days relative to T_obs = max(Date) + 1, Frequency (distinct visits), and cumulative Monetary spend.
- FR-06: ML Clustering & Scoring:** Standardizes features via log1p and StandardScaler, fits K-Means, and assigns 1–5 quantile scores mapping to 8 business tiers.

Non-Functional Requirements (NFR)

- NFR-01 (Latency):** Processes 15,000+ orders in <5.0 seconds using vectorized Pandas C-routines.
- NFR-02 (Reproducibility):** Fixed seed (seed=42) guarantees 100% deterministic clustering across platforms.
- NFR-03 (Modularity):** Clean star schema data marts (FACT_TRANSACTIONS, DIM_CUSTOMERS).

Table 1.1: System Environment Matrix

- Hardware / CPU:** Apple Silicon M-Series / Intel Core i5/i7 (8-Core+)
- System Memory:** 8 GB to 16 GB Unified Memory (Min: 4 GB)
- Storage Drive:** Solid State Drive (NVMe SSD)
- Operating System:** macOS Darwin (Unix POSIX) / Linux / Win10+
- Runtime & Stack:** Python 3.9+, Modern Web Browser, VS Code

Tools & Technology Stack (Multi-Layer Architecture)

1. DATA REFINERY & INGESTION LAYER — Python 3.9+ • Pandas (v2.3.3) • NumPy (v2.0.2)

High-throughput tabular ingestion, schema typing, null elimination, reverse logistics isolation, and vectorized Tukey IQR outlier calculations.

2. MACHINE LEARNING & STATISTICAL MODELING — Scikit-Learn (v1.6.1) • SciPy (v1.13.1)

Log1p transformation for Pareto skew compression, StandardScaler normalization, K-Means clustering with k-means++ centroid initialization, and Silhouette score diagnostics.

3. DIMENSIONAL STORAGE & STAR SCHEMA — Analytical Star Schema Data Marts • CSV / Parquet

Decoupled fact and dimension storage (FACT_TRANSACTIONS, FACT_REVERSALS, DIM_CUSTOMERS) designed for BI compatibility with Tableau and Power BI.

4. EXECUTIVE BI DASHBOARD & ACTIVATION — HTML5 • Tailwind CSS • Chart.js 4.x • Git & GitHub

Client-side canvas graphing engine rendering revenue donuts, spatial RFM bubble charts, KPI scorecards, and searchable CRM activation grids.

Level 0 Data Flow Diagram (Context Architecture)

Figure 2.1: Level 0 Data Flow Diagram (Context-Level DFD)

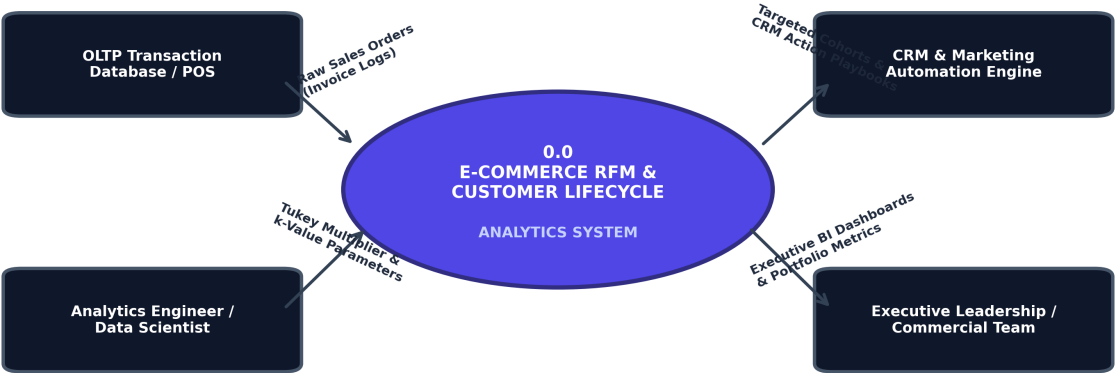


Figure 2.1: Level 0 Context Data Flow Diagram (Macro Boundary)

Context Boundary & Entity Interfaces

- **Process 0.0 (Global Boundary)**: Represents the complete 'E-Commerce Customer Lifecycle & RFM Analytics System' interacting with 4 primary external entities.
- **1. OLTP Sales Stream**: Provides high-frequency transaction logs containing InvoiceNo, StockCode, Quantity, InvoiceDate, UnitPrice, and CustomerID.
- **2. Data Scientist**: Supplies IQR outlier tuning multipliers, cluster count hyperparameters (k in $[2..6]$), and evaluates validation diagnostics.
- **3. CRM Marketing Engine**: Consumes segmented customer lists ('Champions', 'At Risk Whales') for programmatic email, SMS, and ad audience targeting.
- **4. Commercial Executives**: Receives interactive executive KPI scorecards, segment revenue distribution, and churn risk summaries.

Level 1 Data Flow Diagram (End-to-End Analytics Pipeline)

Figure 2.2: Level 1 Data Flow Diagram (End-to-End Analytics Pipeline)

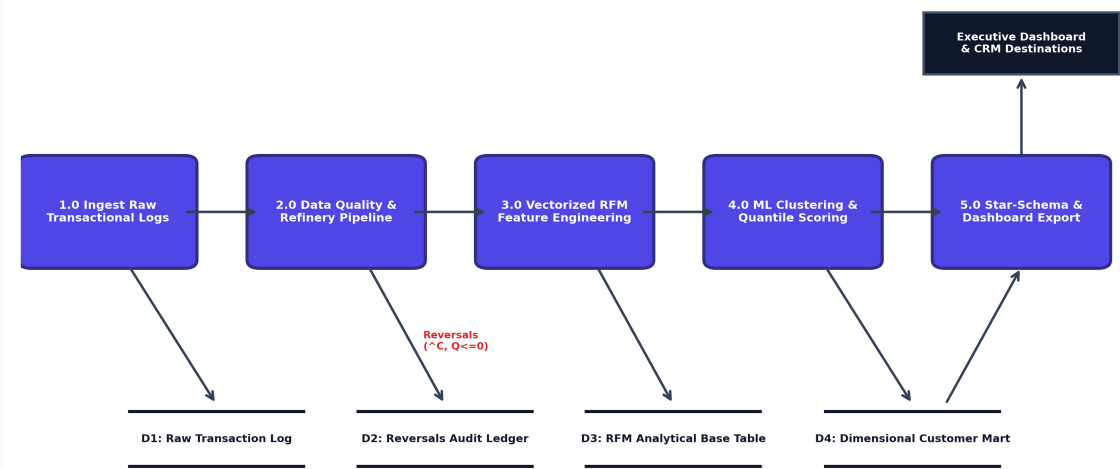


Figure 2.2: Level 1 Data Flow Diagram (Pipeline Architecture)

5 Sequential Pipeline Transformations

- **Process 1.0 (Ingestion & Schema Typing):** Parses raw streams, validates typing, and logs records to D1 (Raw Transaction Log).
- **Process 2.0 (Data Quality & Refinery):** Drops guest IDs, routes returns to D2 (Reversals Audit Ledger), and applies Tukey IQR fences.
- **Process 3.0 (Vectorized RFM Feature Extraction):** Computes Recency, Frequency, Spend, and AOV, writing to D3 (RFM Analytical Base Table).
- **Process 4.0 (ML Clustering & Quantile Scoring):** Executes log1p scaling, StandardScaler, K-Means, and 1–5 scoring into D4 (Segmented Customer Mart).
- **Process 5.0 (Star Schema & Dashboard):** Publishes BI data marts and powers the interactive executive web dashboard.

Level 2 Data Flow Diagram (Refinery & ML Decomposition)

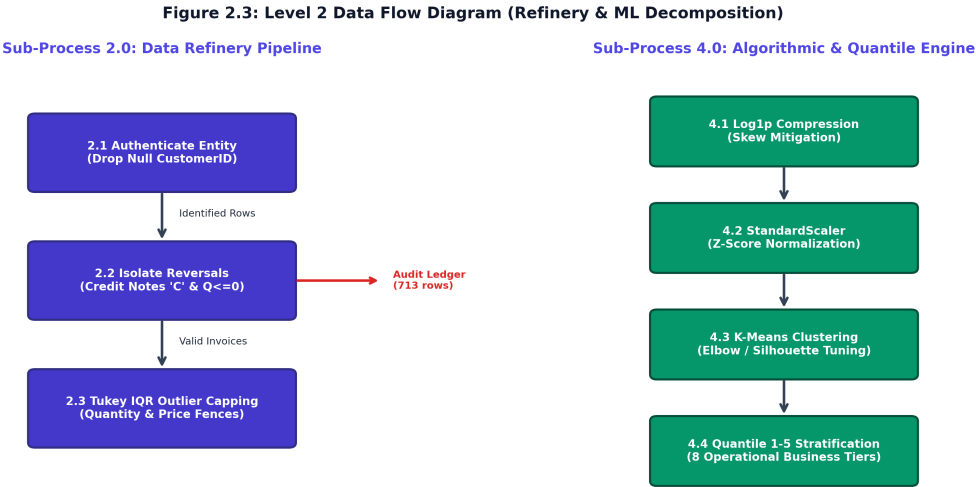


Figure 2.3: Level 2 Data Flow Diagram (Detailed Decomposition)

Algorithmic & Mathematical Decomposition

- **Sub-Process 2.1 (Null ID Purge):** Validates CustomerID existence; drops unauthenticated guest sessions.
- **Sub-Process 2.2 (Reversals Filter):** Detects cancellation identifiers ('^C' or negative Quantity) and exports to fact_reversals_audit.csv.
- **Sub-Process 2.3 (Tukey IQR Capping):** Computes $[Q1 - 1.5 \cdot IQR, Q3 + 1.5 \cdot IQR]$ to eliminate wholesale bulk distributor anomalies.
- **Sub-Process 4.1 (Log1p Scaling):** Applies $z = \ln(1 + x)$ to compress severe right-skewed Pareto customer spending.
- **Sub-Process 4.2 (StandardScaler):** Transforms features to zero mean and unit variance for distance-based clustering.
- **Sub-Process 4.3 (K-Means & Silhouette):** Fits K-Means across k in $[2..6]$, diagnosing optimal clusters via Silhouette coefficient.

UML Use Case Diagram & Actor Responsibilities

Figure 2.4: UML Use Case Diagram (System Actors & Functional Capabilities)

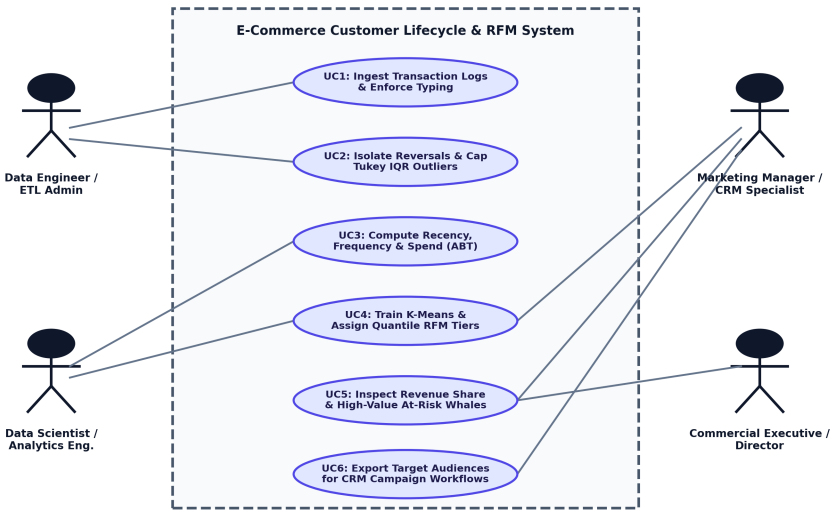


Figure 2.4: UML Use Case Diagram (Actors & Capabilities)

Table 2.1: Use Case Specification Matrix (UC-01 to UC-06)

- UC-01: Ingest Transaction Stream [Data Engineer / Admin]:** Ingests raw event logs, enforces schema typing, verifies datatypes.
- UC-02: Isolate Reversals & IQR Filter [Data Engineer / Admin]:** Purges null IDs, isolates returns (^C), caps Tukey bulk outliers.
- UC-03: Compute Vectorized RFM [Analytics Engineer]:** Aggregates transactions into customer Recency, Frequency, and Spend.
- UC-04: Train & Tune K-Means Model [Data Scientist]:** Applies log1p + StandardScaler, evaluates Elbow/Silhouette, fits k=4.
- UC-05: Inspect Executive KPI Metrics [Commercial Executive]:** Monitors portfolio health (\$65k spend), Champions share (72.5%), at-risk capital.
- UC-06: Export Targeted CRM Audiences [Marketing Manager]:** Filters segments in activation ledger and exports audience lists for campaigns.

Milestone Evaluation Schedule & Project Roadmap

Formal Assessment Schedule & Deliverable Tracking

- **[24/09/2026] First Progress Work Evaluation (10 Marks) — COMPLETED (Current Submission)**
Scope: Sections 1.1, 1.2, 1.3, 2.1, 2.2
- **[13/10/2026] Mid-Term Examination (Second Progress) (20 Marks) — Scheduled for Next Milestone**
Scope: 2.3 (Class Diagram), 2.4 (Sequence Diagrams), 3.1 (Data Dictionary), 4.1 (UI Layouts)
- **[27/10/2026] Third Progress Work Evaluation (10 Marks) — Scheduled for Third Milestone**
Scope: 4.2 (Sample Coding), 4.3 (Navigation), 5.1-5.2 (Conclusion), 6.1 (References)
- **[06/11/2026] Report Submission (Spiral Binding) (10 Marks) — Scheduled for Final Submission**
Scope: Final Signed Spiral Copy & Project Media
- **[18-20/11/2026] Final Examination (SEE) (50 Marks) — Scheduled for Semester Examination**
Scope: Project Execution Demonstration & Viva Voce

MILESTONE STATUS SUMMARY (24/09/2026): All deliverables for the First Progress Work milestone (10 Marks) have been successfully engineered, verified, and compiled into the 10-page report and interactive demonstration dashbo



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THANK YOU!

Questions & Technical Discussion

E-COMMERCE CUSTOMER LIFECYCLE & RFM SEGMENTATION DASHBOARD

Candidate: Mr. Deep Koshiya (B.Sc. in AI & Data Science, Semester V)

Institution: Institute of Advanced Research (IAR), Gandhinagar

Assessment Milestone: First Progress Work (24/09/2026 • 10 Marks)

Course & Project: Project - 1 (CE-502 / PRJ-701)

GitHub Repository: <https://github.com/Deep-k-coder/ecommerce-rfm-analytics>

Project Status: Refinery Pipeline, RFM Engine, UML/DFD Models & Dashboard Complete